

Question ID: 8f0c82e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

The minimum value of x is **12** less than **6** times another number n . Which inequality shows the possible values of x ?

Answer

- A. $x \leq 6n - 12$
- B. $x \geq 6n - 12$
- C. $x \leq 12 - 6n$
- D. $x \geq 12 - 6n$

Correct Answer: B

Rationale

Choice B is correct. It's given that the minimum value of x is **12** less than **6** times another number n . Therefore, the possible values of x are all greater than or equal to the value of **12** less than **6** times n . The value of **6** times n is given by the expression **$6n$** . The value of **12** less than **$6n$** is given by the expression **$6n - 12$** . Therefore, the possible values of x are all greater than or equal to **$6n - 12$** . This can be shown by the inequality **$x \geq 6n - 12$** .

Choice A is incorrect. This inequality shows the possible values of x if the maximum, not the minimum, value of x is **12** less than **6** times n .

Choice C is incorrect. This inequality shows the possible values of x if the maximum, not the minimum, value of x is **6** times n less than **12**, not **12** less than **6** times n .

Choice D is incorrect. This inequality shows the possible values of x if the minimum value of x is **6** times n less than **12**, not **12** less than **6** times n .